INTERNCIRCLE CYBERSECURITY TRACK • PROJECT 05

Linux Permissions and Hardening Lab Report

Analyst: THRIGUNA G.A | **Environment:** Isolated Ubuntu 24.04 LTS VirtualBox VM | **Focus:** Access Control, Service Triage, UFW & SSH Hardening

1. Lab Setup, Scope & Snapshot Baseline

Rules of Engagement & Isolation: All commands and configuration modifications in this lab were executed strictly inside an isolated Oracle VirtualBox virtual machine installed locally on my own workstation. Before performing system updates, service changes, or SSH daemon modifications, a clean baseline snapshot (`Snap01_Clean_Install`) was captured to guarantee instant, risk-free rollback.

Lab Parameter	Configuration & Verification Details
Hypervisor & Guest OS	Oracle VirtualBox 7.1 • Ubuntu Server 24.04.1 LTS (64-bit)
Allocated Resources	2 vCPUs, 2048 MB RAM, 20 GB Dynamically Allocated VDI Storage
Network Adapter Mode	NAT + Host-Only Adapter (`192.168.56.101`) for isolated local SSH testing
Recovery Snapshots	`Snap01_Clean_Install` (pre-hardening) & `Snap02_Post_UFW` (pre-SSH edit)

2. System Package Updates & Patching

Following snapshot creation, the local APT package index was refreshed and all installed system packages were upgraded to remediate known baseline vulnerabilities.

TERMINAL CAPTURE 01 — SYSTEM UPDATE & KERNEL/SECURITY PATCHING thriguna@ubuntu-lab:~

```
thriguna@ubuntu-lab:~$ sudo apt update && sudo apt upgrade -y Hit:1
http://archive.ubuntu.com/ubuntu noble InRelease Get:2
http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB] Reading package
lists... Done Building dependency tree... Done The following packages will be upgraded:
base-files libssl3t64 linux-firmware openssh-client openssh-server systemd udev 7
upgraded, 0 newly installed, 0 to remove and 0 not upgraded. Setting up
libssl3t64:amd64 (3.0.13-0ubuntu3.4) ... Setting up openssh-server (1:9.6p1-
3ubuntu13.5) ... Processing triggers for initramfs-tools ... Done. System packages up
to date.
```

Result Recorded: Seven core security packages—including `libssl3t64`, `openssh-server`, `systemd`, and `linux-firmware`—were patched to current Ubuntu Security Notice levels.

3. Users, Groups & File Permissions Lab

To enforce the principle of least privilege, a dedicated unprivileged account (`student`) and collaborative group (`lab`) were created. Four permission modes—600, 640, 755, and 770—were configured and tested across user contexts.

TERMINAL CAPTURE 02 — USER/GROUP PROVISIONING & PERMISSION VERIFICATION — **thriguna@ubuntu-lab:~/lab_files**

```
      # 1. Create group 'lab', user 'student', and assign group membership
thriguna@ubuntu-lab:~$ sudo groupadd lab thriguna@ubuntu-lab:~$ sudo adduser --
disabled-password --gecos "" student thriguna@ubuntu-lab:~$ sudo usermod -aG lab
student thriguna@ubuntu-lab:~$ groups student student : student lab # 2. Create test
files and apply 600, 640, and 755 permissions thriguna@ubuntu-lab:~$ mkdir -p
~/lab_files && cd ~/lab_files thriguna@ubuntu-lab:~/lab_files$ echo "Top Secret API
Key" > private.txt thriguna@ubuntu-lab:~/lab_files$ echo "Team Lab Schedule" >
group_notes.txt thriguna@ubuntu-lab:~/lab_files$ printf '#!/bin/bash\necho "Health
check OK"\n' > check.sh thriguna@ubuntu-lab:~/lab_files$ chmod 600 private.txt
thriguna@ubuntu-lab:~/lab_files$ sudo chown thriguna:lab group_notes.txt && chmod 640
group_notes.txt thriguna@ubuntu-lab:~/lab_files$ chmod 755 check.sh # 3. Configure
shared group directory (/srv/labshare) with 770 thriguna@ubuntu-lab:~/lab_files$ sudo
mkdir -p /srv/labshare thriguna@ubuntu-lab:~/lab_files$ sudo chown root:lab
/srv/labshare && sudo chmod 770 /srv/labshare # 4. Inspect resulting permission strings
thriguna@ubuntu-lab:~/lab_files$ ls -la -rwxr-xr-x 1 thriguna thriguna 36 Nov 12 19:12
check.sh -rw-r----- 1 thriguna lab 18 Nov 12 19:12 group_notes.txt -rw------- 1
thriguna thriguna 19 Nov 12 19:12 private.txt thriguna@ubuntu-lab:~/lab_files$ ls -ld
/srv/labshare drwxrwx--- 2 root lab 4096 Nov 12 19:13 /srv/labshare # 5. Cross-user
access validation as 'student' thriguna@ubuntu-lab:~/lab_files$ sudo -u student cat
/home/thriguna/lab_files/private.txt cat: /home/thriguna/lab_files/private.txt:
Permission denied thriguna@ubuntu-lab:~/lab_files$ sudo -u student cat
/home/thriguna/lab_files/group_notes.txt Team Lab Schedule thriguna@ubuntu-
lab:~/lab_files$ sudo -u student /home/thriguna/lab_files/check.sh Health check OK
```

Target Object	Octal / Symbolic	Owner : Group	Tested Access Behavior
`private.txt`	600 (`-rw-------`)	`thriguna:thriguna`	Owner can read/write; `student` received *Permission denied*.
`group_notes.txt`	640 (`-rw-r-----`)	`thriguna:lab`	Owner can read/write; `student` (in `lab`) can read only; others blocked.
`check.sh`	755 (`-rwxr-xr-x`)	`thriguna:thriguna`	Owner has full control; `student` and others can read and execute script.
`/srv/labshare`	770 (`drwxrwx---`)	`root:lab`	`root` and `lab` members can create/delete files; non-members have zero access.

4. Service Inventory & Attack Surface Reduction

Active systemd units were enumerated using `systemctl list-units --type=service --state=running` to identify unnecessary background listeners that increase the server's attack surface.

Service Unit	Initial State	Action Taken	Security Rationale
`ssh.service`	Active (running)	Retained & Hardened	Required for remote administration; restricted via key auth and UFW.
`systemd-timesyncd`	Active (running)	Retained	Essential for accurate NTP clock synchronization in security logs.
`cups.service`	Active (running)	Stopped & Disabled	Print spooler unnecessary on a headless server; eliminates network print exposure.
`avahi-daemon.service`	Active (running)	Stopped & Disabled	mDNS/DNS-SD discovery broadcasts host info on local LAN; disabled via `systemctl disable --now`.

5. Host Firewall Configuration (UFW)

Uncomplicated Firewall (UFW) was configured with a strict default-deny inbound policy, permitting only incoming OpenSSH traffic before enabling the firewall.

TERMINAL CAPTURE 03 — UFW POLICY ENFORCEMENT & VERBOSE STATUS thriguna@ubuntu-lab:~

```
thriguna@ubuntu-lab:~$ sudo ufw default deny incoming Default incoming policy changed to 'deny' thriguna@ubuntu-lab:~$ sudo ufw default allow outgoing Default outgoing policy changed to 'allow' thriguna@ubuntu-lab:~$ sudo ufw allow OpenSSH Rule added Rule added (v6) thriguna@ubuntu-lab:~$ sudo ufw --force enable Firewall is active and enabled on system startup thriguna@ubuntu-lab:~$ sudo ufw status verbose Status: active Logging: on (low) Default: deny (incoming), allow (outgoing), disabled (routed) New profiles: skip To Action From -- ------ ---- 22/tcp (OpenSSH) ALLOW IN Anywhere 22/tcp (OpenSSH (v6)) ALLOW IN Anywhere (v6)
```

6. SSH Daemon Hardening & Safe Dual-Session Testing

> **Safe Change Control Protocol:** Before editing `/etc/ssh/sshd_config`, a second active SSH terminal session was kept open as a fallback lifeline, an Ed25519 SSH key pair was deployed to `~/.ssh/authorized_keys` (chmod 600), and the configuration syntax was validated with `sudo sshd -t` prior to reloading.

TERMINAL CAPTURE 04 — SSHD HARDENING & EFFECTIVE PARAMETER AUDIT thriguna@ubuntu-lab:~

```
      # Verified key-based login first, then applied hardened drop-in config
thriguna@ubuntu-lab:~$ sudo tee /etc/ssh/sshd_config.d/99-hardening.conf << 'EOF'
PermitRootLogin no PasswordAuthentication no PubkeyAuthentication yes
PermitEmptyPasswords no MaxAuthTries 3 X11Forwarding no EOF # Syntax check and reload
without dropping existing fallback session thriguna@ubuntu-lab:~$ sudo sshd -t && sudo
systemctl reload ssh # Verify effective active sshd runtime parameters thriguna@ubuntu-
lab:~$ sudo sshd -T | grep -E
"permitrootlogin|passwordauthentication|pubkeyauthentication|maxauthtries"
permitrootlogin no passwordauthentication no pubkeyauthentication yes maxauthtries 3
```

Verification Result: Opening a new test connection confirmed that SSH login succeeded cleanly using the Ed25519 private key, whereas root login attempts and password-only login attempts were immediately rejected with `Permission denied (publickey)`.

7. Key Lessons Learned

- **Why `chmod 777` Must Never Be Used:** Granting read, write, and execute permissions to all local users (`rwxrwxrwx`) allows any compromised service account or unprivileged user to alter scripts, inject malicious code, or delete critical files. Always apply the tightest octal mode required (such as `600` for secrets, `640` for group files, and `750`/`755` for executables).
- **Why UFW Rule Ordering Prevents Lockouts:** Allowing OpenSSH before running `ufw enable` ensures existing and new administrative sessions remain reachable when the default-deny inbound policy activates.
- **Safe Remote Configuration Discipline:** Keeping a secondary active terminal session open and validating syntax with `sshd -t` before reloading prevents accidental administrative lockout if a configuration typo occurs.
- **Defense-in-Depth Synergy:** Combining timely APT security patching, minimal running services, least-privilege file permissions, a default-deny host firewall, and key-only SSH authentication eliminates the most common Linux server attack vectors.